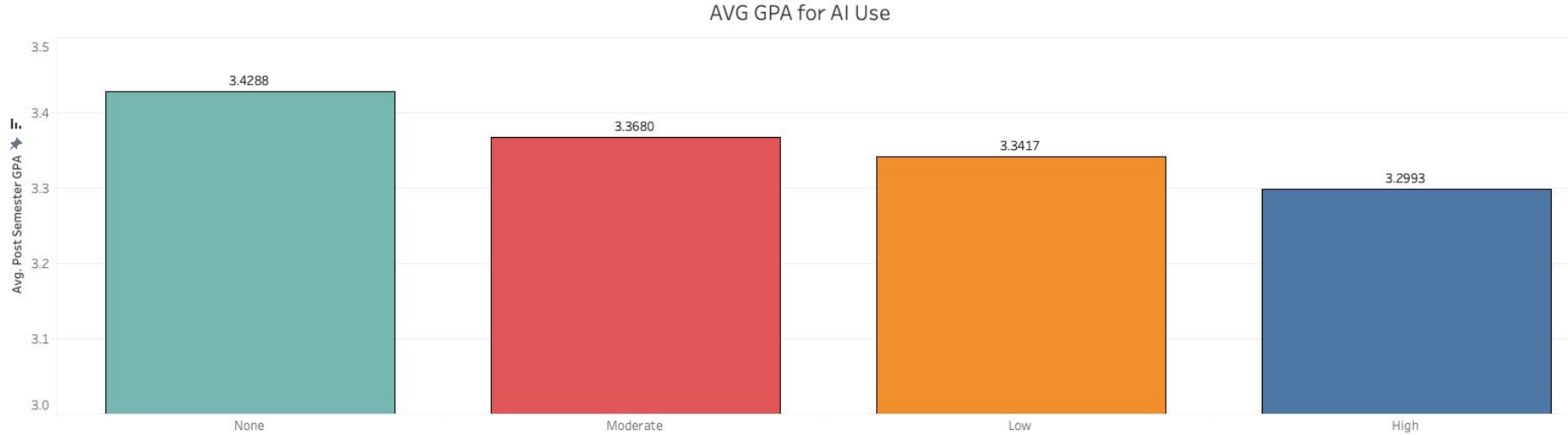


AI Impact on Students

By Dee Christen

A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.

Students with no AI use had the highest average GPA



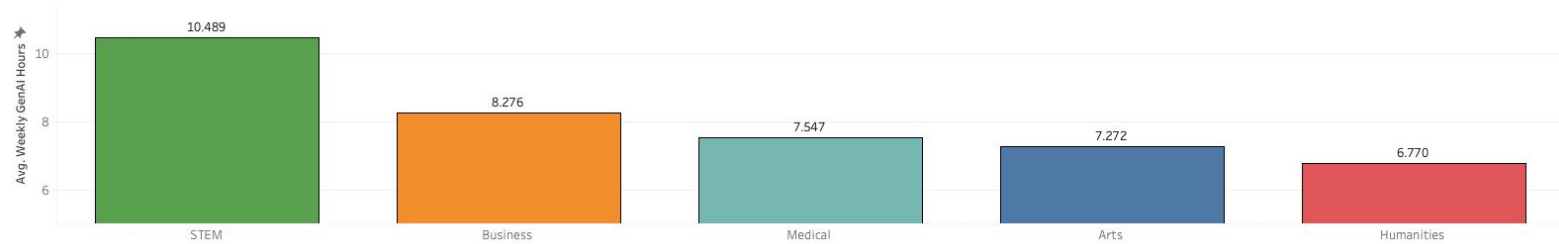
More AI use did not translate into higher GPAs.

Traditional study habits were linked to better performance and skill retention.

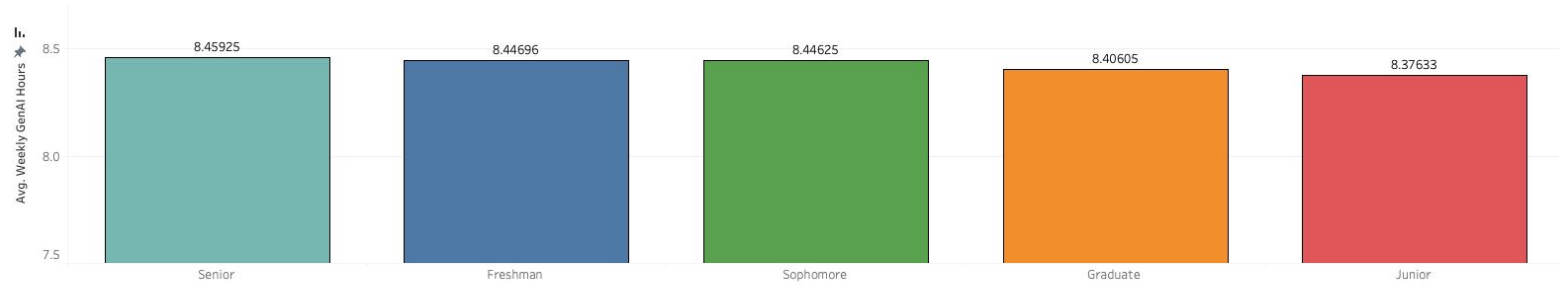
Students relying heavily on AI reported more burnout and anxiety.

Balance—not dependence—appears to produce the best outcomes.

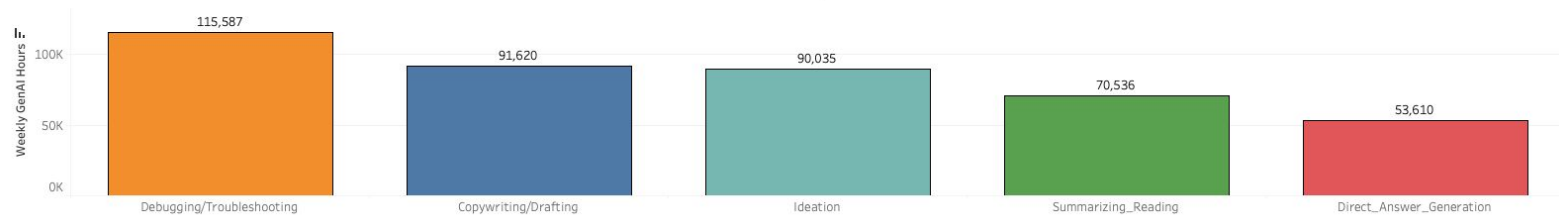
AVG AI Hours per Week for Each Major



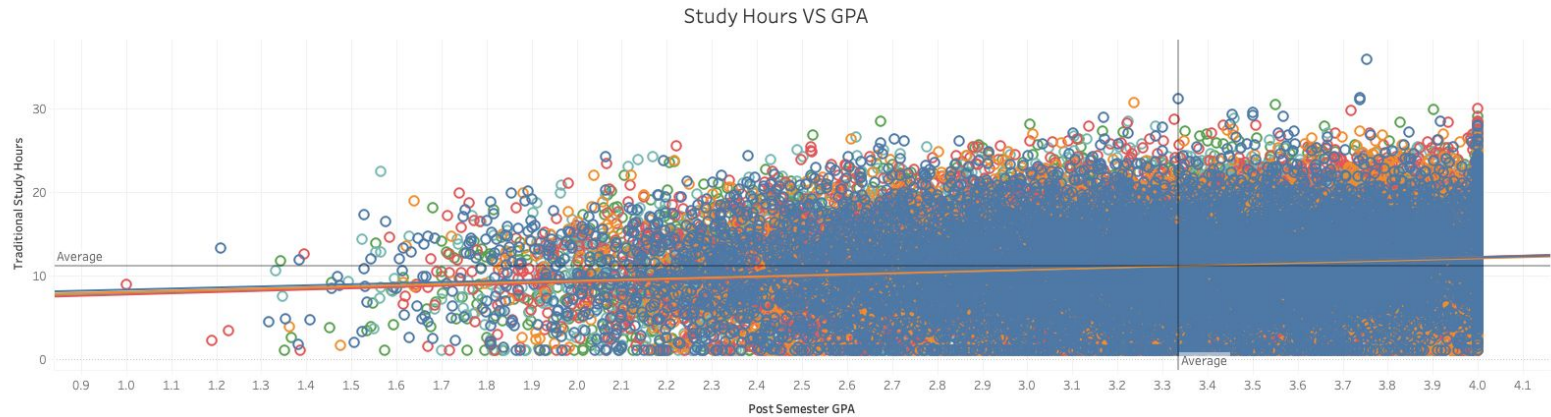
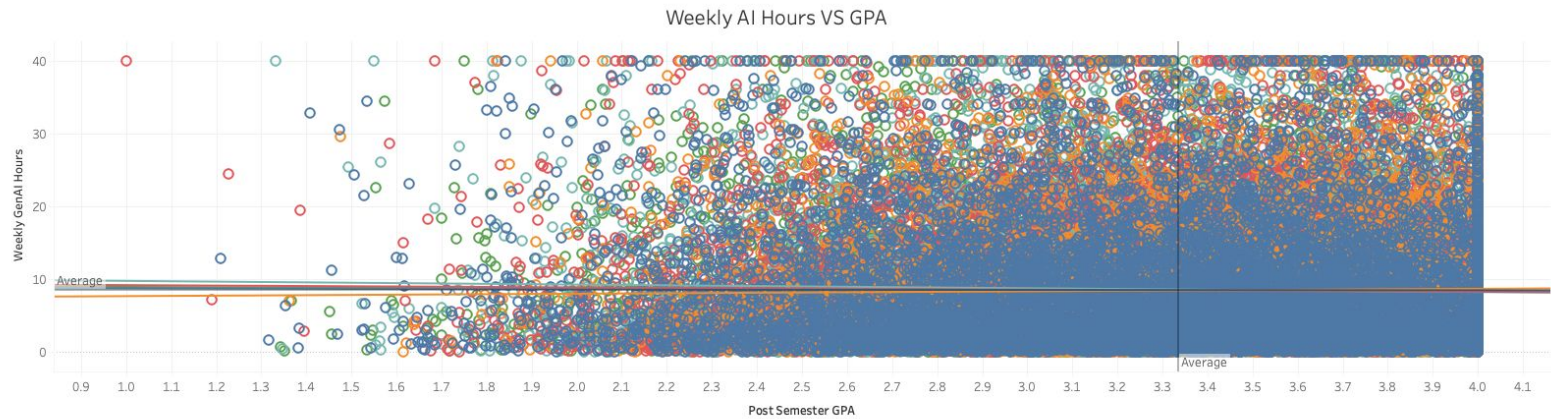
AVG AI Hours per Week for Each Year



AI Use and Primary Use Case

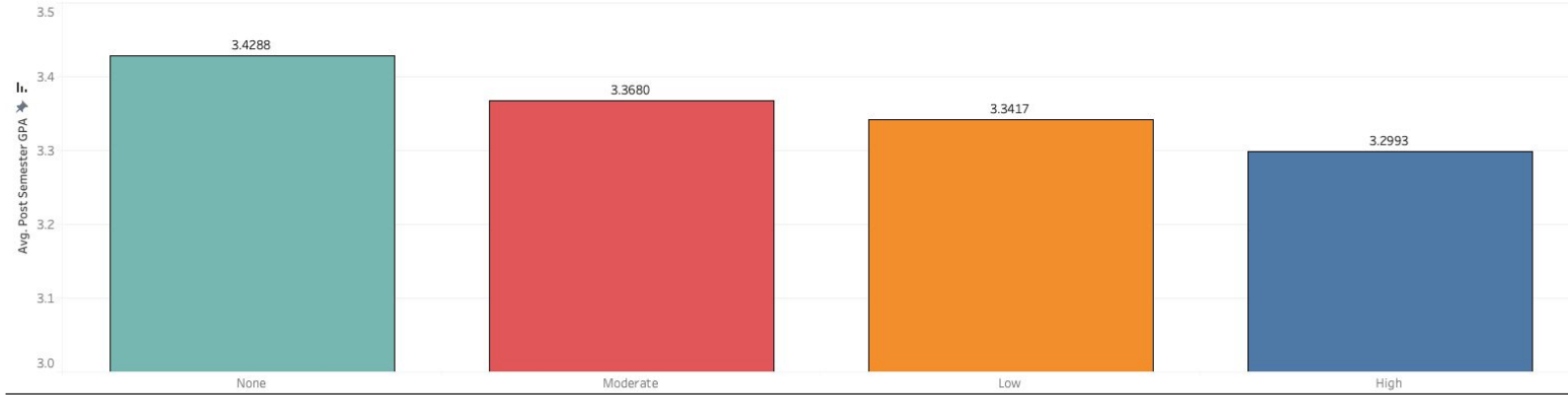


I found that STEM majors used AI the most with Debugging & troubleshooting being the primary use and there was no meaningful difference between years.

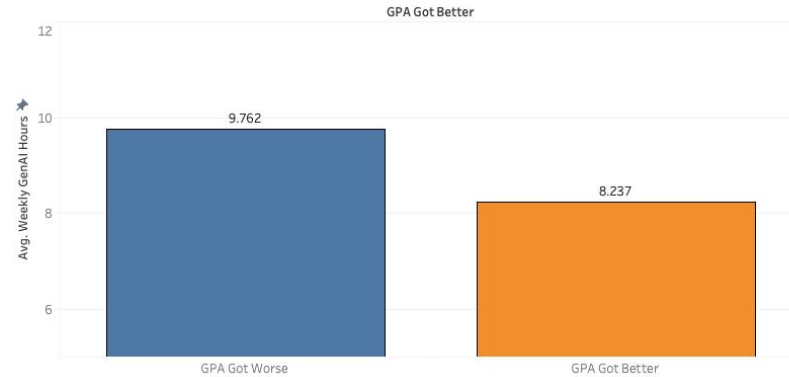


I looked at correlation between AI use and GPA and traditional study and GPA and found that AI had a slight negative correlation while traditional study had a slight positive.

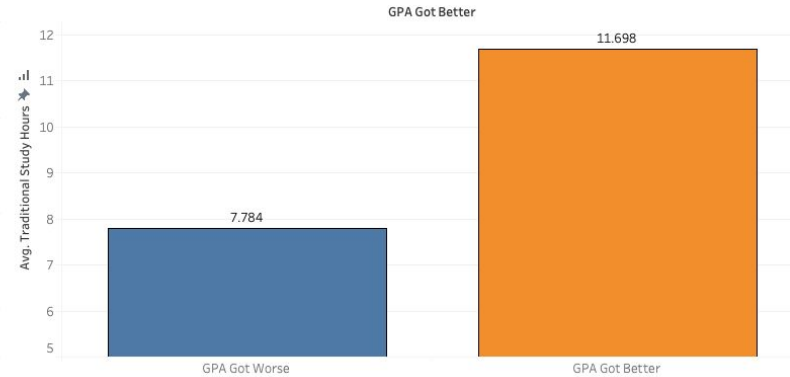
AVG GPA for AI Use



AVG AI Use for people who's GPA got Better

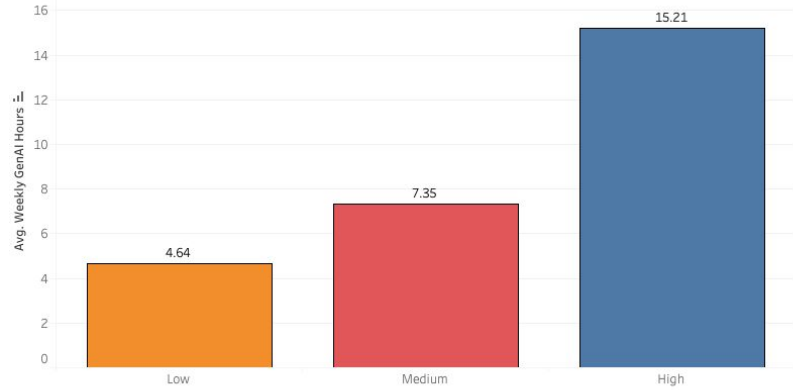


AVG Traditional Study for people who's GPA got Better

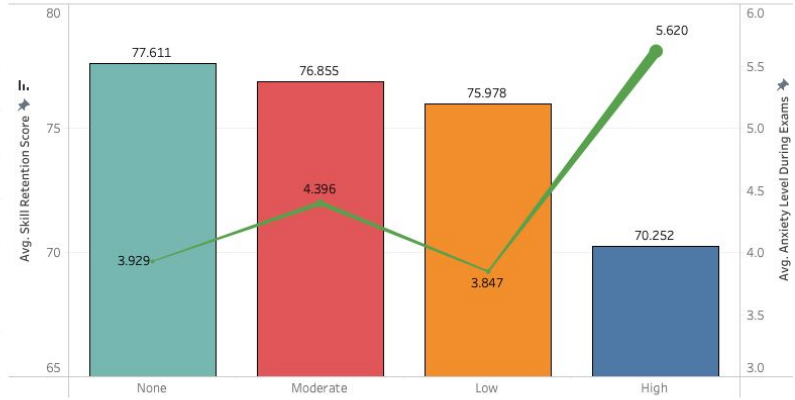


Students who used AI had a lower GPA on average than those who did not use it. Also students whose GPA improved through the semester used less AI and more traditional study hours.

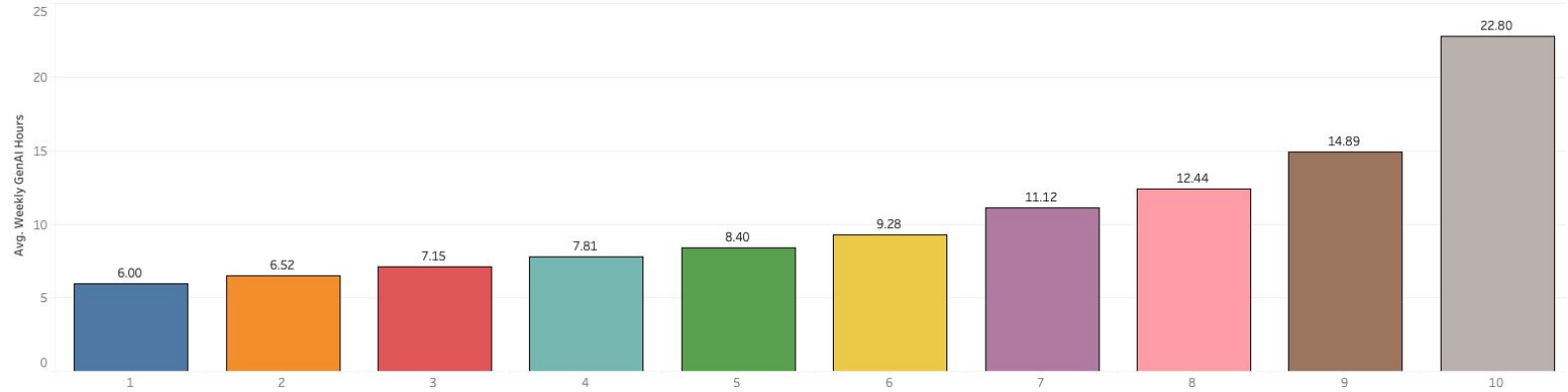
Burnout Risk With AVG AI Use



AI Use With AVG Skill Retention And AVG Anxiety Level



Anxiety Level During Exams with AVG AI Use

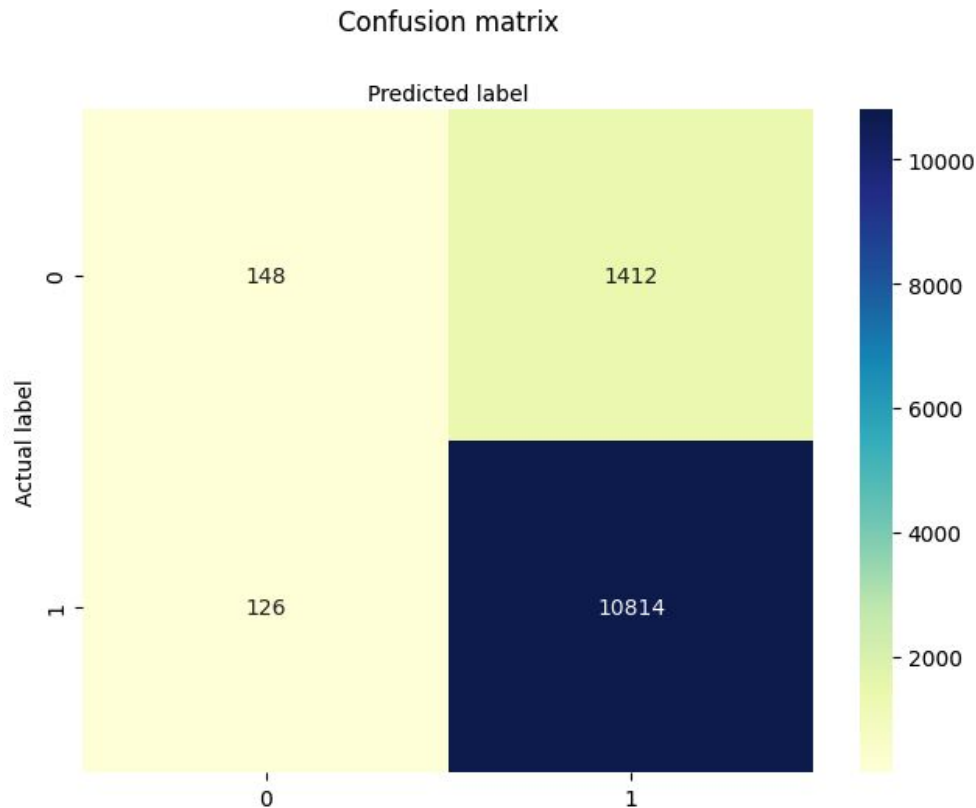


More AI use lead to higher burnout, higher anxiety, and lower skill retention.

Model Evaluation:
Mean Squared Error (MSE): 0.03
Root Mean Squared Error (MSE): 0.16
R-squared (R2): 0.90

Accuracy: 0.877
Precision: 0.8845084246687388
Recall: 0.9884826325411334
F1 Score: 0.9336096002762669

Classification Report:				
	precision	recall	f1-score	support
0	0.54	0.09	0.16	1560
1	0.88	0.99	0.93	10940
accuracy			0.88	12500
macro avg	0.71	0.54	0.55	12500
weighted avg	0.84	0.88	0.84	12500



I ran a linear and logistic regression models. The logistic model was good at getting true positives but not negatives.

Conclusion

AI usage did not significantly improve students GPAs and traditional study helped more. Students that improved their GPA used less AI and more traditional study. Heavy AI users experienced more burnout, more anxiety, and less skill retention. While AI can help in school it works best as a supplement not a replacement.